

Linear Mixed Model (선형혼합모형)

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Today's Topics

Linear Mixed Model

- Fixed Component Inference
- Variance Component Inference

Fixed Component Inference

Inference for the fixed effects

Wald test - chi-square distribution

F test - F-distribution

LR test - chi-square distribution

Fixed Component Inference (Cont'd)

Wald tests

Estimate for β :

$$\hat{\beta}(\alpha) = \left(\sum_{i=1}^N X_i' W_i X_i \right)^{-1} \sum_{i=1}^N X_i' W_i y_i, \quad W_i = V_i^{-1}$$

with α replaced by its ML or REML estimate

Conditional on α , $\hat{\beta}(\alpha)$ is multivariate normal with mean β and covariance

$$\begin{aligned} \text{var}(\hat{\beta}) &= \left(\sum_{i=1}^N X_i' W_i X_i \right)^{-1} \left(\sum_{i=1}^N X_i' W_i \text{var}(Y_i) W_i X_i \right) \left(\sum_{i=1}^N X_i' W_i X_i \right)^{-1} \\ &= \left(\sum_{i=1}^N X_i' W_i X_i \right)^{-1} \end{aligned}$$

Fixed Component Inference (Cont'd)

Wald tests

For any known matrix L , considering testing

$$H_0 : L\beta = 0 \text{ vs } H_A : L\beta \neq 0$$

Wald test statistic:

$$G = \hat{\beta}^t L^t \left[L \left(\sum_{i=1}^N X_i^t V_i^{-1}(\hat{\alpha}) X_i \right)^{-1} L^t \right]^{-1} L \hat{\beta}$$

Asymptotic null distribution of G is χ^2 with $\text{rank}(L)$ degrees of freedom

Fixed Component Inference (Cont'd)

F tests

Wald test based on

$$\text{var}(\hat{\beta}) = \left(\sum_{i=1}^N X_i' W_i(\alpha) X_i \right)^{-1}$$

Variability introduced from replacing α by some estimate is not taken into account in Wald tests

Therefore, Wald tests will only provide valid inferences in sufficiently large samples

In practice, this is often resolved by replacing the χ^2 distribution by an appropriate F-distribution

Fixed Component Inference (Cont'd)

F tests

For any known matrix $\mathbf{L}$, considering testing

$$\mathbf{H}_0 : \mathbf{L}\boldsymbol{\beta} = \mathbf{0} \text{ vs } \mathbf{H}_A : \mathbf{L}\boldsymbol{\beta} \neq \mathbf{0}$$

F test statistic:

$$F = \frac{\hat{\boldsymbol{\beta}}^t \mathbf{L}^t \left[\mathbf{L} \left(\sum_{i=1}^N \mathbf{X}_i^t \mathbf{V}_i^{-1}(\hat{\boldsymbol{\alpha}}) \mathbf{X}_i \right)^{-1} \mathbf{L}^t \right]^{-1} \mathbf{L} \hat{\boldsymbol{\beta}}}{\text{rank}(\mathbf{L})}$$

Approximate null-distribution of F is F with numerator degrees of freedom equal to rank(L)

Fixed Component Inference (Cont'd)

F tests

Denominator degrees of freedom to be estimated from the data

Satterthwaite approximation

Kenward and Roger approximation

In the context of longitudinal data, all methods typically lead to large numbers of degrees of freedom, and therefore also to very similar p-values

For univariate hypothesis ($\text{rank}(L)=1$), the F-test reduces to a t-test

Fixed Component Inference (Cont'd)

Robust inference

Estimate for β :

$$\hat{\beta}(\alpha) = \left(\sum_{i=1}^N X_i' W_i X_i \right)^{-1} \sum_{i=1}^N X_i' W_i y_i, \quad W_i = V_i^{-1}$$

with α replaced by its ML or REML estimate

Conditional on α , β has mean

$$E[\hat{\beta}(\alpha)] = \left(\sum_{i=1}^N X_i' W_i X_i \right)^{-1} \sum_{i=1}^N X_i' W_i E(Y_i) = \beta$$

provided that $E(Y_i) = X_i \beta$

Hence, in order for β to be unbiased, it is sufficient that the mean of the response is correctly specified

Fixed Component Inference (Cont'd)

Robust inference

Conditional on α , $\hat{\beta}$ has covariance

$$\begin{aligned}\text{var}(\hat{\beta}) &= \left(\sum_{i=1}^N X_i' W_i X_i \right)^{-1} \left(\sum_{i=1}^N X_i' W_i \text{var}(Y_i) W_i X_i \right) \left(\sum_{i=1}^N X_i' W_i X_i \right)^{-1} \\ &= \left(\sum_{i=1}^N X_i' W_i X_i \right)^{-1}\end{aligned}$$

if $\text{var}(Y_i)$ is correctly specified

Fixed Component Inference (Cont'd)

Robust inference

Note that this assumes that the covariance matrix $\text{var}(Y_i)$ is correctly modeled as $V_i = Z_i D Z_i^t + R_i$

This covariance estimate is therefore often called the 'naïve' estimate

The so-called 'robust' estimate for $\text{var}(\hat{\beta})$ which does not assume the covariance matrix to be correctly specified is obtained from replacing $\text{var}(Y_i)$ by $[Y_i - X_i \hat{\beta}][Y_i - X_i \hat{\beta}]^t$ rather than V_i

The only condition for $[Y_i - X_i \hat{\beta}][Y_i - X_i \hat{\beta}]^t$ to be unbiased for $\text{var}(Y_i)$ is that the mean is again correctly specified

Fixed Component Inference (Cont'd)

Robust inference

The obtained estimate is called the 'robust' variance estimate, also called the sandwich estimate:

$$\text{var}(\hat{\beta}) = \underbrace{\left(\sum_{i=1}^N X_i' W_i X_i \right)^{-1}}_{\text{BREAD}} \underbrace{\left(\sum_{i=1}^N X_i' W_i \text{var}(Y_i) W_i X_i \right)}_{\text{MEAT}} \underbrace{\left(\sum_{i=1}^N X_i' W_i X_i \right)^{-1}}_{\text{BREAD}}$$

For highly unbalanced data, the type-1 error rate cannot be controlled

Fixed Component Inference (Cont'd)

Robust inference

Based on this sandwich estimate, robust versions of the Wald test as well as of the approximate t-test and F-test can be obtained

Note that this suggests that as long as interest is only in inferences for the mean structure, little effort should be spent in modeling the covariance structure, provided that the data set is sufficiently large

Fixed Component Inference (Cont'd)

Robust inference

Extreme point of view: OLS with robust standard errors

Appropriate covariance modeling may still be of interest:

- For the interpretation of random variation in data

- For gaining efficiency

In presence of missing data, robust inference is valid under very severe assumptions about the underlying missingness process

Fixed Component Inference (Cont'd)

LR tests

Comparison of nested models with different mean structures, but equal covariance structure

Null hypothesis of interest equals $H_0: \beta \in \Theta_{\beta,0}$, for some subspace $\Theta_{\beta,0}$ of the parameter space Θ_{β} of the fixed effects β

Fixed Component Inference (Cont'd)

LR tests

Notation:

L_{ML} : ML likelihood function

$\hat{\theta}_{ML,0}$: MLE under H_0

$\hat{\theta}_{ML}$: MLE under general model

Test statistic:

$$-2 \ln \lambda_N = -2 \ln \left[\frac{L_{ML}(\hat{\theta}_{ML,0})}{L_{ML}(\hat{\theta}_{ML})} \right],$$

Asymptotic null distribution: χ^2 with df equals to the difference in dimension of Θ_β and $\Theta_{\beta,0}$

Fixed Component Inference (Cont'd)

LR tests

Under REML, the response Y is transformed into error contrasts $U = A^t Y$, for some matrix A with $A^t X = 0$

Afterwards, ML estimation is performed based on the error contrasts

Models with different mean structures lead to different sets of error contrasts

Hence, the corresponding REML likelihoods are based on different observations, which makes them no longer comparable

Conclusion: LR tests for the mean structure are not valid under REML

Variance Component Inference

Inference for the variance components

Wald test

LR test

Variance Component Inference (Cont'd)

Inference for the variance components

Inference for the mean structure is usually of primary interest

However, inferences for the covariance structure is of interest as well:

- Interpretation of the random variation in the data

- Overparameterized covariance structures lead to inefficient inferences for mean structure

- Too restrictive models invalidate inferences for the mean structure

Variance Component Inference (Cont'd)

Wald test

Asymptotically, ML and REML estimates of α are normally distributed with corrected mean and inverse Fisher information matrix as covariance

Hence approximate s.e's and Wald tests can easily be obtained

Variance Component Inference (Cont'd)

Boundary Problem

The quality of the normal approximation for the ML or REML estimates strongly depends on the true value α

Poor normal approximation if α is relatively close to the boundary of the parameter space

If α is a boundary value, the normal approximation completely fails

Variance Component Inference (Cont'd)

Boundary Problem

One of the Wald tests for the variance components presents

$$\mathbf{H}_0 : \mathbf{d}_{33} = \mathbf{0}$$

However the null hypothesis automatically indicates $d_{13} = d_{23} = 0$

Hence, the test is only correct, when the null-hypothesis is not a boundary value (e.g., $d_{33} = 0.1$)

Note that, even under the hierarchical model interpretation, a classical Wald test is valid for testing $H_0: d_{23} = 0$

Variance Component Inference (Cont'd)

LR test

Comparison of nested models with equal mean structures, but different covariance structure

Null hypothesis of interest equals $H_0: \alpha \in \Theta_{\alpha,0}$, for some subspace $\Theta_{\alpha,0}$ of the parameter space Θ_α of the variance components α

Variance Component Inference (Cont'd)

LR test

Notation:

L_{ML} : ML likelihood function

$\hat{\theta}_{ML,0}$: MLE under H_0

$\hat{\theta}_{ML}$: MLE under general model

Test statistic:

$$-2 \ln \lambda_N = -2 \ln \left[\frac{L_{ML}(\hat{\theta}_{ML,0})}{L_{ML}(\hat{\theta}_{ML})} \right],$$

Asymptotic null distribution: χ^2 with df equals to the difference in dimension of Θ_α and $\Theta_{\alpha,0}$

Variance Component Inference (Cont'd)

LR test

Note that, as long as models are compared with the same mean structure, a valid LR test can be obtained under REML as well

Indeed, both models can be fitted using the same error contrasts, making the likelihoods comparable

Note that, if H_0 is a boundary value, the classical χ^2 approximation may not be valid

For some very specific null-hypotheses on the boundary, the correct asymptotic null-distribution has been derived

Variance Component Inference (Cont'd)

LR test

Under a hierarchical model interpretation, the asymptotic null distribution for the LR test statistic for testing significance of all variance components related to one or multiple random effects, can be derived

Example: testing whether the variance components associated to the quadratic random time effect are equal to zero, is equivalent to testing

$$H_0 : \mathbf{d}_{13} = \mathbf{d}_{23} = \mathbf{d}_{33} = \mathbf{0}$$

Note that, under the hierarchical interpretation of the model, H_0 is on the boundary of the parameter space

Variance Component Inference (Cont'd)

No random effects versus one random effect

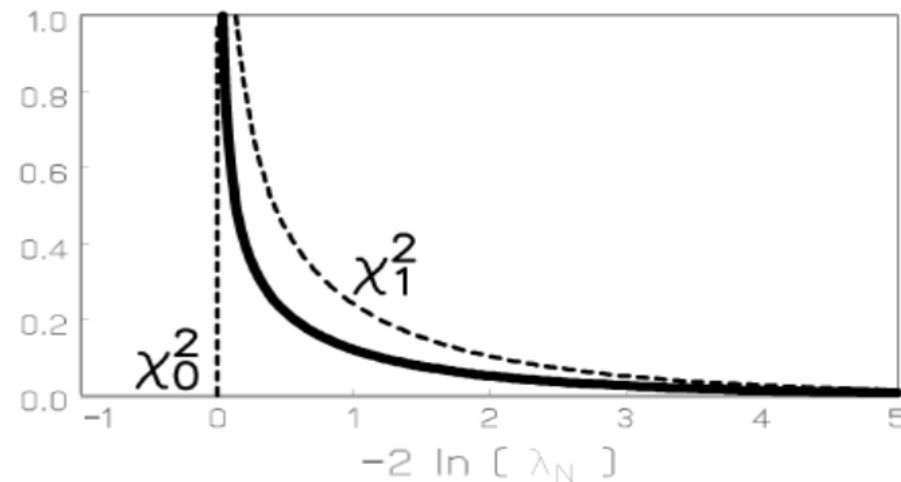
Hypothesis of interest:

$$H_0 : D = 0 \text{ vs } H_A : D = d_{11}$$

for some non-negative scalar d_{11}

Asymptotic null-distribution: $-2\ln\lambda_N \rightarrow \chi^2_{0:1}$

Mixture of χ^2_0 and χ^2_1 with equal weights 0.5



Variance Component Inference (Cont'd)

No random effects versus one random effect

Under H_0 , $-2\ln\lambda_N$ equals 0 in 50% of the cases

Intuitive explanation:

consider the extended parameter space $\mathbb{R}$ for d_{11}

under H_0 , $\hat{d}_{11}$ will be negative in 50% of the cases

under the restriction $d_{11} \geq 0$, these cases lead to $\hat{d}_{11} = 0$

hence, $L_{ML}(\hat{\theta}_{ML,0}) = L_{ML}(\hat{\theta}_{ML})$ in 50% of the cases

Variance Component Inference (Cont'd)

One versus two random effect

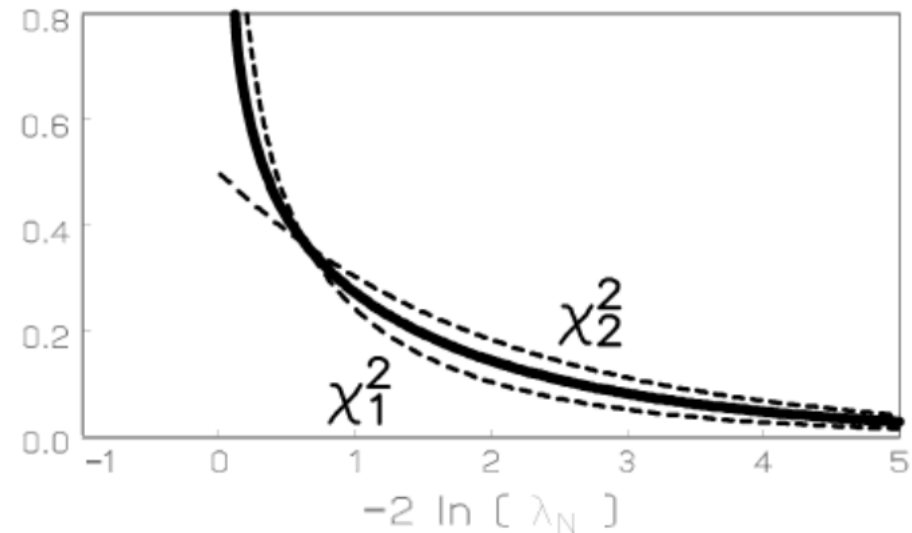
Hypothesis of interest:

$$H_0 : D = \begin{pmatrix} d_{11} & 0 \\ 0 & 0 \end{pmatrix}, \text{ vs } H_1 : \text{not } H_0$$

for $d_{11} > 0$, versus H_1 that D is (2×2) positive semidefinite

Asymptotic null-distribution: $-2\ln\lambda_N \rightarrow \chi_{1:2}^2$

Mixture of χ_1^2 and χ_2^2 with equal weights 0.5



Variance Component Inference (Cont'd)

q versus $q + 1$ random effect

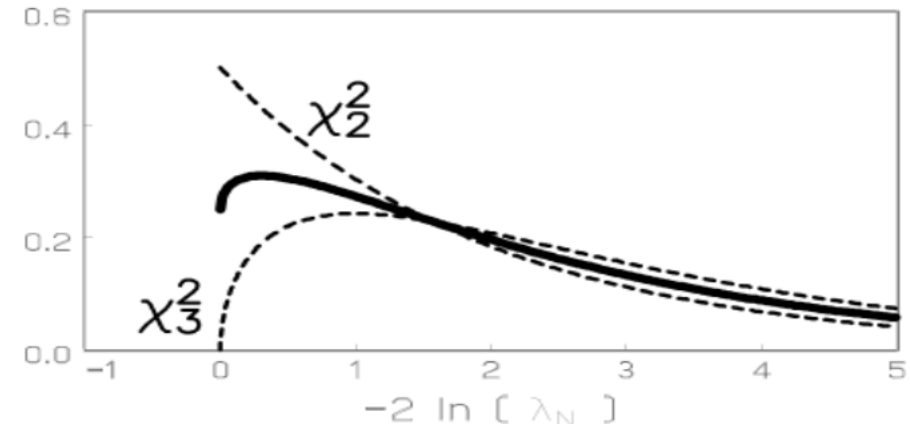
Hypothesis of interest:

$$H_0 : D = \begin{pmatrix} D_{11} & 0 \\ 0' & 0 \end{pmatrix}, \text{ vs } H_1 : \text{not } H_0$$

for D_{11} ($q \times q$) positive definite, versus H_1 that D is $(q + 1) \times (q + 1)$ positive semidefinite

Asymptotic null-distribution: $-2\ln\lambda_N \rightarrow \chi^2_{q:q+1}$

Mixture of χ^2_q and χ^2_{q+1} with equal weights 0.5



Variance Component Inference (Cont'd)

q versus $q + k$ random effect

Hypothesis of interest:

$$H_0 : D = \begin{pmatrix} D_{11} & 0 \\ 0' & 0 \end{pmatrix}, \text{ vs } H_1 : \text{not } H_0$$

for D_{11} ($q \times q$) positive definite, versus H_1 that D is $(q + k) \times (q + k)$ positive semidefinite

Simulations are required to derive asymptotic null distribution

Variance Component Inference (Cont'd)

Conclusions

Correcting for the boundary problem reduces p-values

Thus, ignoring the boundary problem too often leads to over-simplified covariance structures

Hence, ignoring the boundary problem may invalidate inferences, even for the mean structure